

PVsyst - Simulation report

Grid-Connected System

Project: **BATTERY (4)**

Variant: **15.12**

No 3D scene defined, no shadings

System power: **15.12** kWp

Kampung Chuang - Malaysia

Author

NORTHERN SOLAR SDN. BHD. (Malaysia)



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PVsyst V8.1.3

VC0, Simulation date:
with V8.1.3

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Project summary

Geographical Site

Kampung Chuang

Malaysia

Situation

Latitude 3.4869 °(N)

Longitude 101.6284 °(E)

Altitude 22 m

Time zone UTC+8

Project settings

Albedo 0.20

Weather data

Kampung Chuang

Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

Simulation for year no 1

No 3D scene defined, no shadings

Orientation #1

Fixed plane

Tilt/Azimuth 20 / 90°

Orientation #4

Fixed plane

Tilt/Azimuth 20 / 90°

Orientation #2

Fixed plane

Tilt/Azimuth 20 / 90°

Near Shadings

no Shadings

Orientation #3

Fixed plane

Tilt/Azimuth 20 / 90°

User's needs

Unlimited load (grid)

System information

PV Array

Nb. of modules

24 units

Pnom total

15.12 kWp

Inverters

Nb. of units

1 unit

Total power

15 kWac

Pnom ratio

1.01

Results summary

Simulation step

Hourly

Produced Energy

21512 kWh/year

Specific production

1423 kWh/kWp/year

Perf. Ratio PR

82.70 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

Simulation step

Hourly

Orientation #1

Fixed plane

Tilt/Azimuth **20 / 90 °**

Orientation #4

Fixed plane

Tilt/Azimuth **20 / 180 °**

Near Shadings

no Shadings

Orientation #2

Fixed plane

Tilt/Azimuth **20 / -90 °**

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

User's needs

Unlimited load (grid)

Orientation #3

Fixed plane

Tilt/Azimuth **20 / 0 °**

Horizon

Average Height 2.6 °

PV Array Characteristics

PV module

Manufacturer **Astronergy**Model **CHSM66RN(DG)/F-BH-630**

(Custom parameters definition)

CHSM66RN(DG)F-BH-630.PANUnit Nom. Power **630 Wp**

Array #1 - PV Array

Orientation **1**
Tilt/Azimuth **20/90 °**
Number of PV modules **6** units
Nominal (STC) **3780 Wp**
Modules 1 unit x **6** In series

At operating cond. (50°C)

Pmpp **3528 Wp**
U mpp **226 V**
I mpp **16 A**

Array #2 - Sub-array #2

Orientation **2**
Tilt/Azimuth **20/-90 °**
Number of PV modules **6** units
Nominal (STC) **3780 Wp**
Modules 1 unit x **6** In series

At operating cond. (50°C)

Pmpp **3528 Wp**
U mpp **226 V**
I mpp **16 A**

Array #3 - Sub-array #3

Orientation **3**
Tilt/Azimuth **20/0 °**
Number of PV modules **6** units
Nominal (STC) **3780 Wp**
Modules 1 unit x **6** In series

At operating cond. (50°C)

Pmpp **3528 Wp**
U mpp **226 V**
I mpp **16 A**

Inverter

Manufacturer **Hoymiles**Model **HIT-15L-G3**

(Custom parameters definition)

Unit Nom. Power **15.0 kWac**Number of inverters 1 * MPPT **25% 0.3** unit
Total power **3.8 kWac**Operating voltage **150-900 V**
Pnom ratio (DC:AC) **1.01**Number of inverters 1 * MPPT **25% 0.3** unit
Total power **3.8 kWac**Operating voltage **150-900 V**
Pnom ratio (DC:AC) **1.01**Number of inverters 1 * MPPT **25% 0.3** unit
Total power **3.8 kWac**Operating voltage **150-900 V**
Pnom ratio (DC:AC) **1.01**

**PV Array Characteristics****Array #4 - Sub-array #4**

Orientation	4		
Tilt/Azimuth	20/180 °		
Number of PV modules	6 units	Number of inverters	1 * MPPT 25% 0.3 unit
Nominal (STC)	3780 Wp	Total power	3.8 kWac
Modules	1 unit x 6 In series		
At operating cond. (50°C)		Operating voltage	150-900 V
Pmpp	3528 Wp	Pnom ratio (DC:AC)	1.01
U mpp	226 V		
I mpp	16 A		
Total PV power		Total inverter power	
Nominal (STC)	15 kWp	Total power	15 kWac
Total	24 modules	Number of inverters	1 unit
Module area	64.8 m²	Pnom ratio	1.01
Cell area	60.5 m²	No power sharing	

Array losses**Array Soiling Losses**

Loss Fraction 1.5 %

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

LID - Light Induced Degradation

Loss Fraction 1.45 %

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 1.00 % at MPP

Module average degradation

Year no 1
Loss factor 0.55 %/year
Imp / Vmp contributions 80% / 20%
Mismatch due to degradation
Imp RMS dispersion 0.4 %/year
Vmp RMS dispersion 0.4 %/year

IAM loss factor

ASHRAE Param.: IAM = 1 - bo (1/cosi -1)
bo Param. 0.05

DC wiring losses

Global wiring resistance 10 mΩ
Loss Fraction 1.71 % at STC

Array #1 - PV Array

Global array res. 471 mΩ
Loss Fraction 3.00 % at STC

Array #3 - Sub-array #3

Global array res. 236 mΩ
Loss Fraction 1.50 % at STC

Array #2 - Sub-array #2

Global array res. 236 mΩ
Loss Fraction 1.50 % at STC

Array #4 - Sub-array #4

Global array res. 236 mΩ
Loss Fraction 1.50 % at STC

System losses**Unavailability of the system**

Time fraction 1.0 %
3.7 days,
3 periods



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AC wiring losses

Inv. output line up to injection point

Inverter voltage **400** Vac tri
Loss Fraction 1.00 % at STC

Global System

Wire section Copper **3 x 3** mm²
Wires length **14** m



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Main results

Simulation step

Hourly

System Production

Produced Energy **21512** kWh/year

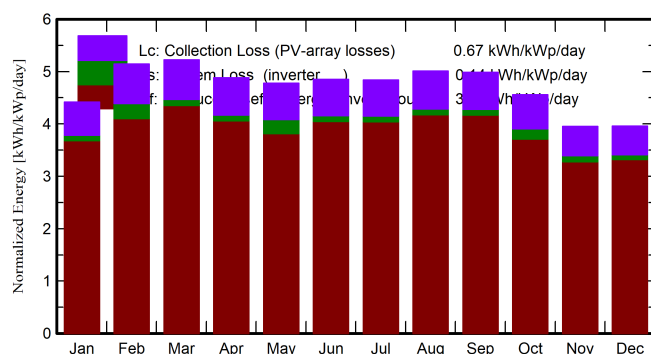
Specific production

1423 kWh/kWp/year

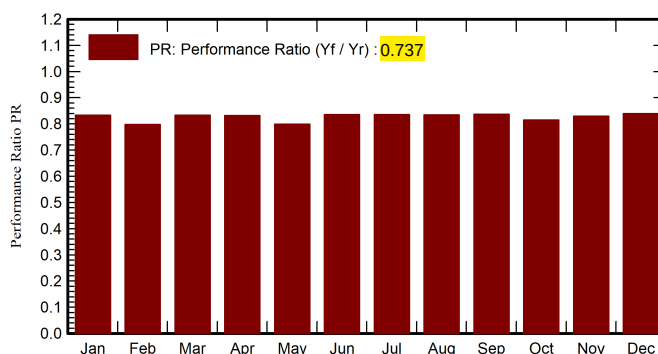
Perf. Ratio PR

82.70 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m ²	DiffHor kWh/m ²	T_Amb °C	GlobInc kWh/m ²	GlobEff kWh/m ²	EArray kWh	E_Grid kWh	PR ratio
January	141.8	70.84	26.85	137.0	129.9	1774	1727	0.834
February	149.0	72.94	27.46	144.0	137.1	1860	1737	0.798
March	167.7	87.92	27.68	162.0	154.5	2097	2042	0.834
April	151.7	74.54	27.73	146.4	139.4	1892	1842	0.832
May	153.4	76.58	28.05	148.2	140.9	1915	1791	0.800
June	150.5	73.96	27.96	145.6	138.4	1887	1839	0.835
July	155.4	74.21	27.56	150.0	142.9	1946	1894	0.835
August	160.8	80.04	27.59	155.2	148.0	2011	1958	0.834
September	154.9	82.48	27.18	149.6	142.5	1943	1894	0.837
October	146.2	81.34	27.07	141.3	134.2	1832	1742	0.815
November	122.9	70.64	26.68	118.6	112.3	1540	1487	0.830
December	127.2	79.83	26.64	122.7	116.2	1601	1558	0.840
Year	1781.5	925.32	27.37	1720.4	1636.3	22297	21512	0.827

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

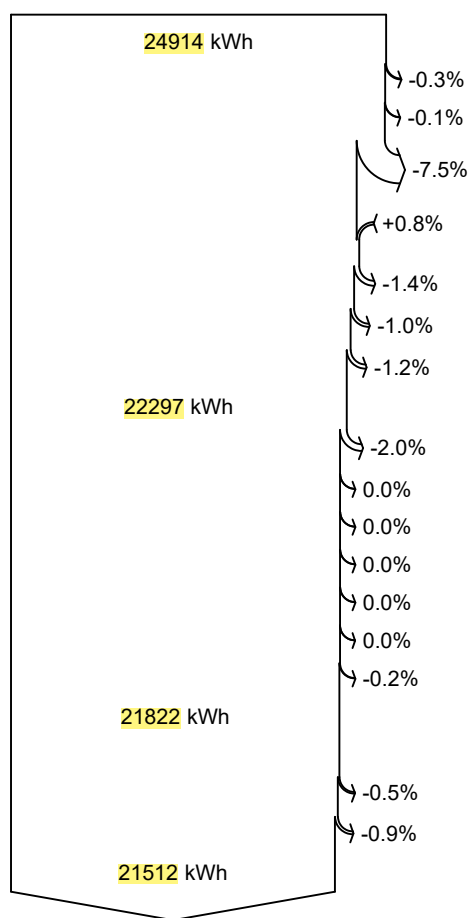
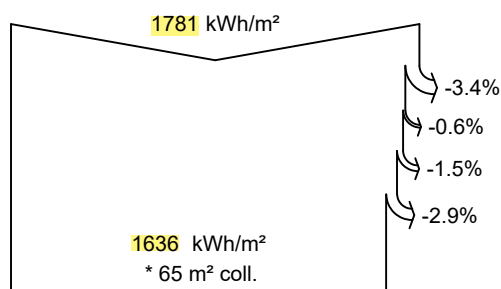
EArray Effective energy at the output of the array

E_Grid Energy injected into grid

PR Performance Ratio



Loss diagram



Global horizontal irradiation

Global incident in coll. plane

Far Shadings / Horizon

Soiling loss factor

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

Module Degradation Loss (for year #1)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

LID - Light induced degradation

Module array mismatch loss

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

AC ohmic loss

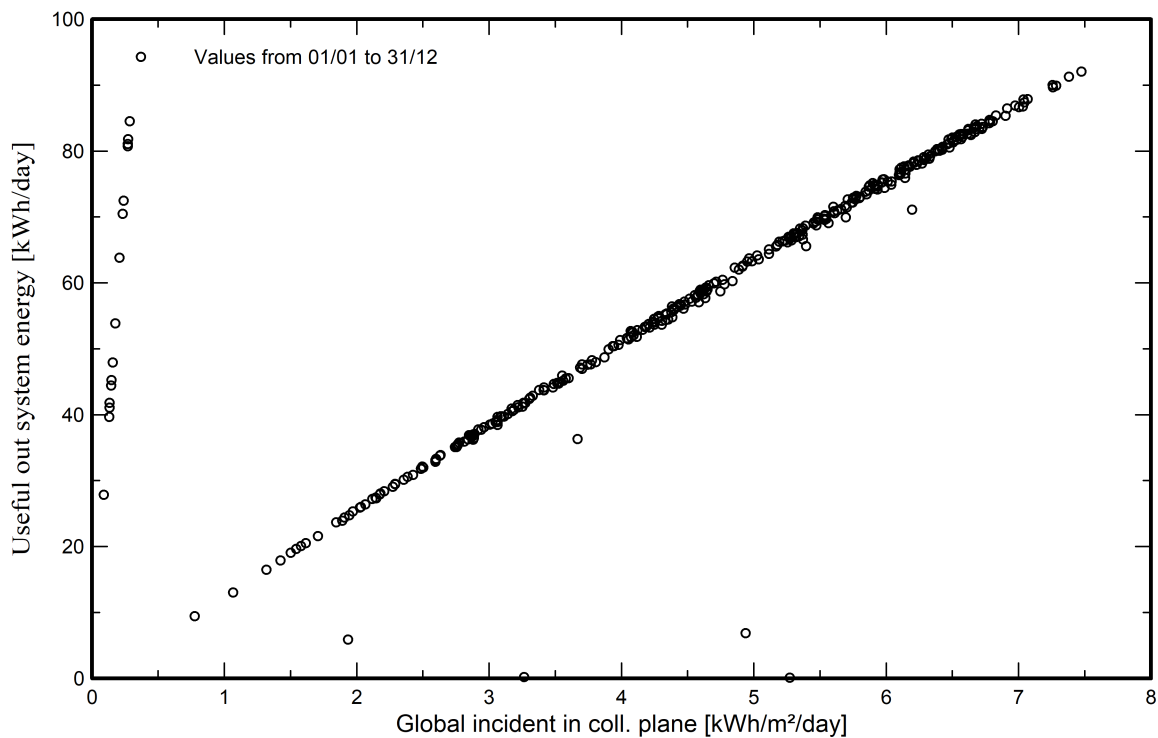
System unavailability

Energy injected into grid

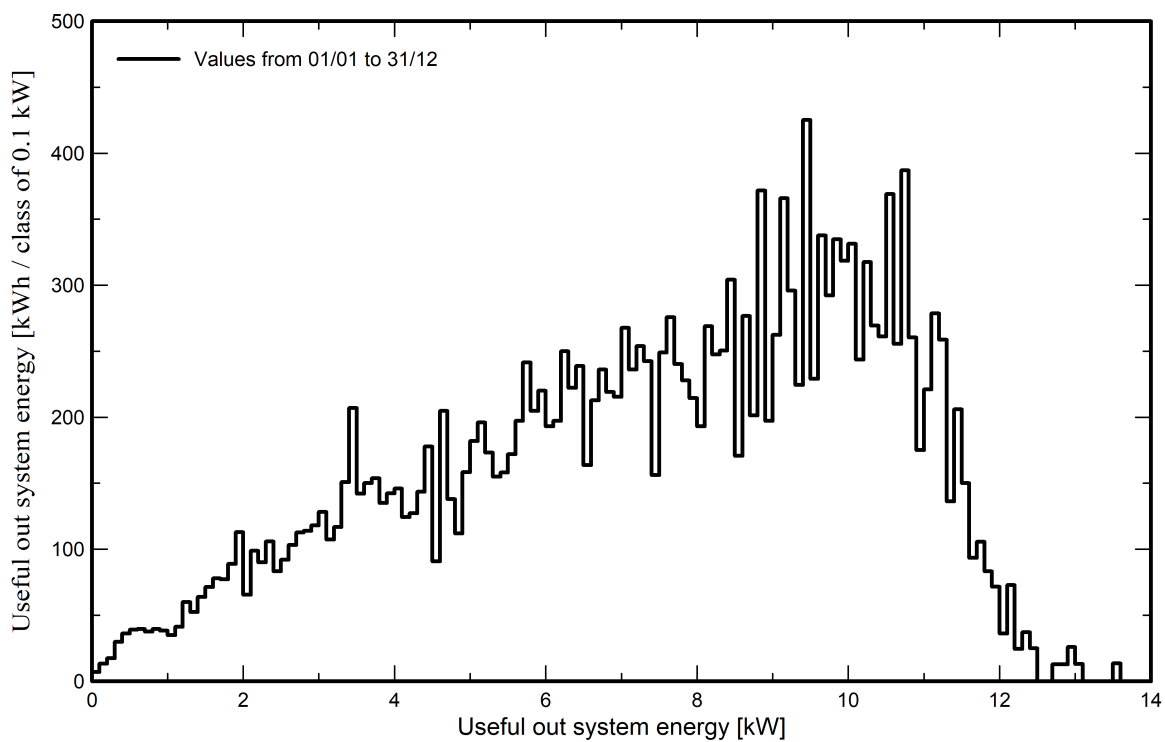


Predef. graphs

Daily Input/Output diagram



System Output Power Distribution





P50 - P90 evaluation

Weather data

Source	Meteonorm 9.0 dll, Sat=100%
Kind	Specific year
Year	Synthetic
Year-to-year variability(Variance)	4.0 %
Specified Deviation	
Year deviation from average	0.0 %

Global variability (weather data + system)

Variability (Quadratic sum)	5.9 %
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Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %
Custom variability	4.0 %

Annual production probability

Variability	666 kWh
P50	11221 kWh
P90	10367 kWh
P75	10772 kWh

Probability distribution

